Reg. No. | | | | | | | | | | | [12CE7604]

SEVENTH SEMESTER B.E EXAMINATION DECEMBER -2015

SOLID WASTE MANAGEMENT

Time:3 Hours **Maximum Marks:70**

Instructions: 1. Answer Any Five Questions

2. Missing data if any may be assumed suitably.

1.	**a.**	Discuss briefly the rate of solid waste generation and the factors associated with it.	**06**
	b.	Explain the term "Garden City to Garbage City" and the impact of solid waste generation on environment.	**08**
2.	**a.**	Give the specifications of the on-site storage methods used to store solid waste and explain the methods.	**06**
	b.	Explain a case study showing the role of Non-Governmental organizations in solid waste Management.	**08**
3.	**a.**	The following Data pertains to a city corporation area collector efficiency = 15Kg/hr and 7 hours effective working per day No. of dwellings = 50000 Assume mean population of 6 residences per dwelling Percapita production of solid waste = 0.4 Kg/day/person Estimate the number of collectors required per day and the quantity of solid waste generated per annum.	**06**
	b.	List the tools and equipments required for primary collection of waste.	**08**
4.	**a.**	Define the term route optimization to transfer waste for offsite processing and give the factors to be considered for selection of the same.	**08**
	b.	Give a neat flow diagram of biogas plant developed by Danish Energy Agency (LINTRUP).	**06**
5.	**a.**	Explain the 4R principle of reduce, Recover, recycle and Reuse in solid waste Management.	**06**
	b.	Write explanatory notes on: **i.** Best Management practices **ii.** Reuse of plastic wastes **iii.** Energy Recovery **iv.** Density Separation.	**08**

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6. a. Give the advantages and disadvantages of sanitary landfill. Illustrate with neat sketches any three methods of sanitary landfill. **08**

b. Design a sanitary Landfill site for the following data. **06**

Population served = 1,00,000

Solid waste generation rate = 2Kg/day/person

Compacted density of solid waste in Land fill = 750 Kg/m^3

Average compacted depth of solid waste = 3m

Assume a rectangular shape with length = twice Breadth

7. a. One ton of solid waste having the formula $C_{50}H_{100}O_{40}N$ is required to be oxidized as per the following equations.

$$C_aH_bO_cN_d + \frac{(4a+b-2C-3d)}{4}O_2 \rightarrow aCO_2 + \frac{b-3d}{2}H_2O + dNH_3$$

$$NH_3 + 2O_2 \rightarrow H_2O + HNO_3$$

Assume air contains 23.15% O_2 by weight and density of air is 1.2928 Kg/m^3

Atomic weights: C=12, H=1, O=16, N=14

Calculate the amount of air required for the above case.

b. Explain the process of Landfill gas movement. **06**

8. a. Explain the treatment methods used for hazardous wastes. **06**

b. Write explanatory notes on **08**

i. E-Waste management.

ii. Integrated solid waste management plan.